WORKED SOLUTION

STEP II 2004, Question 12

Probability and Statistics • Continuous Distributions

STEP II 2004, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q12

The key idea

Everything flows from one antiderivative: xe^{-ax^2} is, up to a constant, the derivative of e^{-ax^2}. That gives the normalising constant and a closed form for the cumulative distribution function, so the median comes from solving one exponential equation. The mode is where the curve peaks, provided that peak lies inside $[0,1]$; otherwise the density is still climbing at $x = 1$. The conditional probability then needs only values of the cumulative distribution function at m and h.

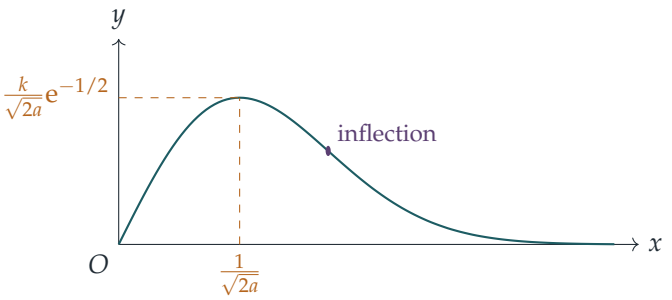
Solution

The sketch

Let $g(x) = kxe^{-ax^2}$ for $x \geq 0$. Then $g(0) = 0$, $g(x) > 0$ for $x > 0$, and by the product rule

$$g'(x) = ke^{-ax^2}(1 - 2ax^2),$$

which is positive for $0 \leq x < 1/\sqrt{2a}$ and negative afterwards. So g rises from the origin (with gradient k there) to a single maximum at $x = 1/\sqrt{2a}$, where $y = \dfrac{k}{\sqrt{2a}}e^{-1/2}$, and then decreases. As $x \to \infty$ the exponential beats the linear factor, so $y \to 0$: the x-axis is an asymptote. (Differentiating again shows a point of inflection at $x = \sqrt{3/(2a)}$, where the curve switches from concave to convex.)



The constant k

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}e^{-ax^2} = -2axe^{-ax^2}$, for $0 \le t \le 1$,

$$\int_0^t kxe^{-ax^2}\,\mathrm{d}x = \left[-\frac{k}{2a}e^{-ax^2}\right]_0^t = \frac{k}{2a}(1-e^{-at^2}).$$

The total probability is 1, so taking $t = 1$ gives $\dfrac{k}{2a}(1-e^{-a}) = 1$, that is,

$$k = \frac{2a}{1-e^{-a}},$$

as required. (The denominator is positive because $a > 0$.) Substituting this back, the cumulative distribution function on $[0,1]$ is

$$F(t) = P(X \le t) = \frac{1-e^{-at^2}}{1-e^{-a}}.$$

The mode

The density is the curve we sketched, cut off at $x = 1$. From the sketch, on $[0,1]$ it is largest at the turning point if that lies in the interval, and otherwise at the right-hand end.

- If $a > \frac{1}{2}$, then $2a > 1$, so $1/\sqrt{2a} < 1$: the peak is inside the interval and $m = \dfrac{1}{\sqrt{2a}}$.
- If $a < \frac{1}{2}$, then $1/\sqrt{2a} > 1$, so $f'(x) > 0$ throughout $[0,1)$: the density is increasing on the whole interval and $m = 1$.

(When $a = \frac{1}{2}$ both formulae give $m = 1$.)

The median

The median h satisfies $F(h) = \frac{1}{2}$:

$$1-e^{-ah^2} = \tfrac{1}{2}(1-e^{-a}) \quad\Longleftrightarrow\quad e^{-ah^2} = \frac{1+e^{-a}}{2}.$$

Taking logarithms and using $h > 0$,

$$h = \sqrt{\frac{1}{a}\ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right)}.$$

This is real since $1+\mathrm{e}^{-a} < 2$, so the logarithm is positive.

When is $h > m$?

If $a \le \frac{1}{2}$ then $m = 1$, and $h < 1$ (half the probability lies below h, and none above 1), so $h < m$. So $h > m$ can happen only when $a > \frac{1}{2}$, and then $m = 1/\sqrt{2a}$. Comparing squares (both are positive),

$$\begin{aligned} h > m &\iff \frac{1}{a}\ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right) > \frac{1}{2a} \iff \ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right) > \frac{1}{2} \\ &\iff \frac{2}{1+\mathrm{e}^{-a}} > \mathrm{e}^{1/2} \iff 1+\mathrm{e}^{-a} < 2\mathrm{e}^{-1/2} \\ &\iff \mathrm{e}^{-a} < 2\mathrm{e}^{-1/2} - 1 \iff a > -\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr). \end{aligned}$$

We multiplied by $a > 0$ in the first step, used that exp is increasing in the third and that ln is increasing in the last. The last step also needs $2\mathrm{e}^{-1/2} - 1 > 0$, which holds because $\mathrm{e}^{1/2} < 2$ (as $\mathrm{e} < 4$).

> **?** **Ask yourself** The chain above assumed $a > \frac{1}{2}$. Doesn't the result need that as an extra condition?

No, and that's why the next part asks for it. We show below that $c = -\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr)$ exceeds $\frac{1}{2}$. So if $a > c$ then automatically $a > \frac{1}{2}$, the formula $m = 1/\sqrt{2a}$ applies, and the chain shows $h > m$. Together with the case $a \le \frac{1}{2}$, this shows that $h > m$ exactly when $a > c$.

The constant exceeds $\frac{1}{2}$

Since ln is increasing,

$$-\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr) > \tfrac{1}{2} \iff 2\mathrm{e}^{-1/2} - 1 < \mathrm{e}^{-1/2} \iff \mathrm{e}^{-1/2} < 1,$$

and the last inequality is true. (Numerically, $c \approx 1.546$.)

The conditional probability

Suppose $a > c$. Then $m < h$, so the event $\{X > m\} \cap \{X < h\}$ is the event $\{m < X < h\}$. Since $P(X < h) = \frac{1}{2}$,

$$P(X > m \mid X < h) = \frac{P(m < X < h)}{P(X < h)} = \frac{F(h) - F(m)}{\frac{1}{2}} = 1 - 2F(m).$$

With $m = 1/\sqrt{2a}$ we have $am^2 = \frac{1}{2}$, so $F(m) = \dfrac{1 - \mathrm{e}^{-1/2}}{1 - \mathrm{e}^{-a}}$. Hence

$$P(X > m \mid X < h) = \frac{(1 - \mathrm{e}^{-a}) - 2(1 - \mathrm{e}^{-1/2})}{1 - \mathrm{e}^{-a}} = \frac{2\mathrm{e}^{-1/2} - \mathrm{e}^{-a} - 1}{1 - \mathrm{e}^{-a}},$$

as required.

Checks

- At $a = c$ the numerator $2\mathrm{e}^{-1/2} - \mathrm{e}^{-a} - 1$ is exactly 0, matching $m = h$ there; it is positive for $a > c$, as a probability should be.
- As $a \to \infty$ the answer tends to $2\mathrm{e}^{-1/2} - 1 \approx 0.213$, which is less than 1.
- With $a = 2.5$: $k \approx 5.447$, $m \approx 0.447$, $h \approx 0.496$, and numerical integration of the density gives $P(X > m \mid X < h) \approx 0.1427$, agreeing with the formula. For $a = 1$ we get $m \approx 0.707 > h \approx 0.616$, consistent with $1 < c$.

Common slips

- Giving $m = 1/\sqrt{2a}$ in every case: when $a < \frac{1}{2}$ the turning point lies beyond $x = 1$, where the density is zero.
- Forgetting that the k inside F must be the normalised value, so that $F(1) = 1$.
- Writing $P(X > m \mid X < h) = P(X > m)/P(X < h)$; the numerator must be the probability of *both* events.

Beyond STEP

This density is a truncated Rayleigh distribution: without the cut-off at $x = 1$, $kx\mathrm{e}^{-ax^2}$ with $k = 2a$ is the distribution of the distance from the origin of a point whose two coordinates are independent normal variables. Comparing mean, median and mode, and how truncation shifts them, is a recurring theme in a first statistics course. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read more about continuous random variables and their density and distribution functions.

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